



Chapter 12

KINETIC THEORY

Class 11 Physics

NEET UG

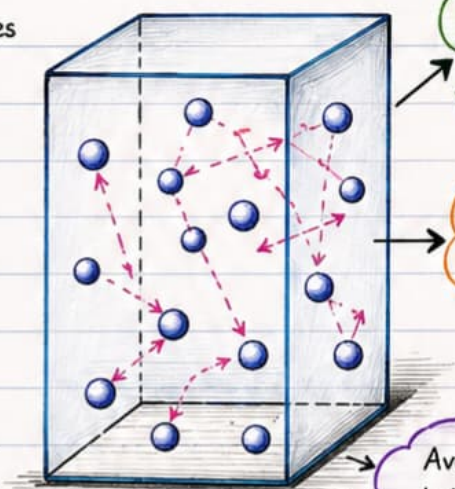


66 The microscopic view of matter and the theory behind pressure, temperature and gas laws. 99

1. INTRODUCTION

Kinetic theory explains the behaviour of gases in terms of the **random motion** of tiny particles called **molecules**.

- Gases consist of a **very large** number of molecules.
- Molecules are in **constant, random** motion.
- Collisions between molecules and with walls are **perfectly elastic**.
- The actual volume of molecules is **negligible** compared to the volume of the gas.



Molecules move randomly in all directions

Collisions are perfectly elastic

Average distance between molecules is much larger than their size

2. ASSUMPTIONS OF KINETIC THEORY

- A gas consists of a **large number** of tiny molecules.
- The molecules are in continuous, **random motion**.
- The actual volume of molecules is **negligible**.
- No **intermolecular forces** act between molecules.
- Collisions between molecules and with walls are **perfectly elastic**.
- The **mean free path** (average distance travelled between collisions) is large compared to the size of molecules.

Key Idea

Gas behaviour arises not from external factors, but from the **motion** and **collisions** of its molecules.



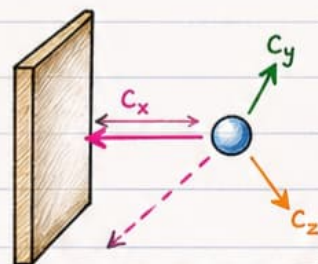
3. PRESSURE OF A GAS (Kinetic Explanation)

Pressure is exerted by gas molecules due to their **collisions** with the walls of the container.

- Consider a molecule of **mass m** hitting the wall with velocity component c_x .
- Change in momentum on collision = $2m c_x$
- If 'z' such molecules hit area A per second, then pressure,

$$P = \frac{1}{3} \frac{Nm \bar{c}^2}{V} = \frac{1}{3} \rho \bar{c}^2$$

N = total number of molecules
 V = volume of gas
 $\bar{c}^2$ = mean square speed
 ρ = density of gas



Only c_x component changes on collision

Note: Pressure is directly proportional to mean square speed of molecules and number density.

Higher temperature $\rightarrow$ Higher speed $\rightarrow$ More pressure

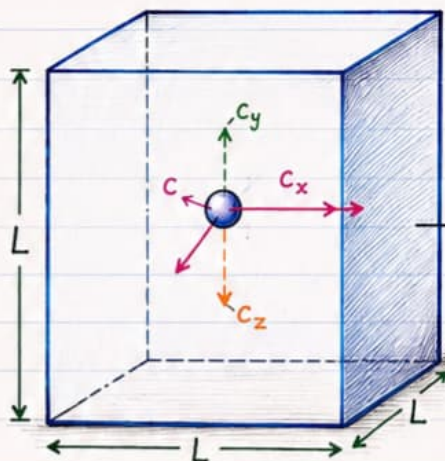
4. DERIVATION OF PRESSURE EXPRESSION

Consider a cubical container of side 'L' containing N molecules.

- A molecule of mass m moving with velocity component c_x hits the wall perpendicular to x -axis.
- Change in momentum in one collision = $2m c_x$
- If it hits the wall A times per second, the force exerted by this molecule on the wall = $2m c_x A$
- Total force on the wall due to all molecules = $2m \sum c_x A$
- Since pressure = Force/Area, and area = L^2

So,

$$\begin{aligned}
 P &= \frac{2m}{L^2} \sum c_x A \\
 &= \frac{2m}{L^2} \left(\frac{N}{L} \right) \overline{c_x^2} \rightarrow A = \frac{\overline{c_x^2}}{L} \quad (\text{From K.T.}) \\
 &= \frac{Nm \overline{c_x^2}}{L^3} \quad (\text{Since } \sum c_x = 0 \text{ but } \sum c_x^2 \neq 0) \\
 \boxed{P = \frac{1}{3} \rho \overline{c^2}} &\quad (\text{Final Pressure Formula})
 \end{aligned}$$



Only the velocity component perpendicular to the wall produces pressure. (Here c_x)

Where,

- N = total number of molecules
- m = mass of one molecule
- L = side of cube
- $\overline{c_x^2}$ = mean square value of velocity component c_x
- $\overline{c^2}$ = mean square speed
- $\overline{c^2} = \overline{c_x^2} + \overline{c_y^2} + \overline{c_z^2}$
- $= 3 \overline{c_x^2}$ (by symmetry)
- $\rho = \frac{Nm}{L^3}$ = density of gas

5. KINETIC ENERGY OF GAS MOLECULES

- Each molecule of mass m moving with speed c has kinetic energy' $\frac{1}{2} m c^2$.
- Average (mean) kinetic energy of a molecule $\rightarrow \frac{1}{2} m \overline{c^2}$
- Total kinetic energy of N molecules $= \frac{1}{2} Nm \overline{c^2}$

IMPORTANT RESULT

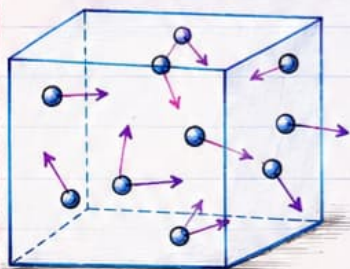
Average kinetic energy of gas molecules depends ONLY on absolute temperature T .

$$\frac{1}{2} m \overline{c^2} = \frac{3}{2} kT$$

Where, k = Boltzmann constant
 $= 1.38 \times 10^{-23} \text{ J K}^{-1}$



3D VIEW (Random Motion)



Molecules move in all directions with different speeds.

Their collisions with walls cause pressure.

SUMMARY

- ★ Gases consist of a very large number of tiny molecules in continuous random motion.
- ★ No intermolecular forces in ideal gas.
- ★ Collisions are perfectly elastic.
- ★ Pressure is due to collisions of molecules with container walls.
- ★ Average kinetic energy $\propto$ Temperature (T).



6. RELATION BETWEEN PRESSURE AND TEMPERATURE

- From the final pressure formula,

$$P = \frac{1}{3} \rho \bar{c}^2$$

- Also, average kinetic energy of a molecule

$$\frac{1}{2} m \bar{c}^2 = \frac{3}{2} kT$$

- So, $\bar{c}^2 = \frac{3kT}{m}$

- Put this value in pressure formula,

$$P = \frac{1}{3} \rho \times \frac{3kT}{m}$$

$$P = \frac{\rho k}{m} T \quad \dots\dots (1)$$

- Thus, pressure of a gas is **directly proportional** to the **absolute temperature**, if volume is constant.
- This is **Gay-Lussac's law**.

DERIVATION OF EQUATION (1)

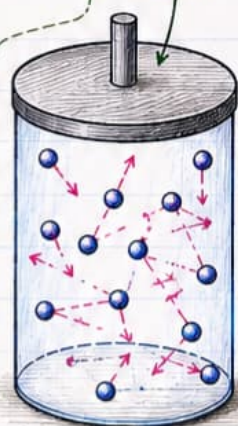
$$P = \frac{1}{3} \rho \bar{c}^2 \quad \text{and} \quad \frac{1}{2} m \bar{c}^2 = \frac{3}{2} kT$$

$$\Rightarrow \bar{c}^2 = \frac{3kT}{m}$$

$$\Rightarrow P = \frac{1}{3} \rho \left(\frac{3kT}{m} \right)$$

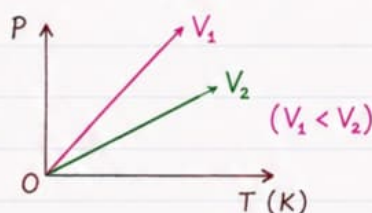
$$P = \frac{\rho k}{m} T$$

Constant Volume (V)



MEANING:

At constant volume, if temperature increases, molecules move faster, collisions increase, so pressure increases.



At constant volume,
 $P \propto T$
Graph is a straight line.

7. ROOT MEAN SQUARE (r.m.s.) SPEED

- Molecules in a gas have **different speeds**.
- Root Mean Square speed c_{rms} is defined as

$$c_{rms} = \sqrt{\frac{c_1^2 + c_2^2 + c_3^2 + \dots + c_N^2}{N}}$$

- For an ideal gas,

$$c_{rms} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$$

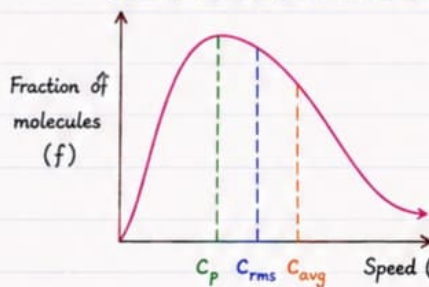
Where,

k = Boltzmann constant ($1.38 \times 10^{-23} \text{ J K}^{-1}$)

R = Gas constant ($8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)

m = mass of one molecule, M = molar mass

SPEED DISTRIBUTION CURVE



In a gas,
 $C_p < c_{rms} < C_{avg}$

C_p = most probable speed

c_{rms} = root mean square speed

C_{avg} = average speed

The area under the curve is 1 (or 100%) since it represents total molecules.

8. COMBINED GAS EQUATION (FROM KINETIC THEORY)

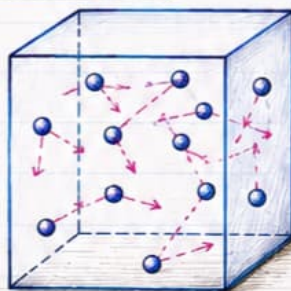
- For n moles of an ideal gas,
Number of molecules = $n N_A$
(N_A = Avogadro's number)

- Using pressure formula,

$$P = \frac{1}{3} \rho \bar{c}^2$$

- After substitution and simplification,

$$PV = nRT \quad \dots\dots (\text{Ideal Gas Equation})$$



GAS CONSTANT R

$$R = N_A k = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$$

CONDITIONS OF IDEAL GAS

- Low pressure
- High temperature
- Molecules are very small compared to separation
- No intermolecular forces

KEY TAKEAWAY:



Gases behave due to random motion of tiny molecules. Collisions with walls cause pressure. Pressure depends on molecular speed, number density and temperature.

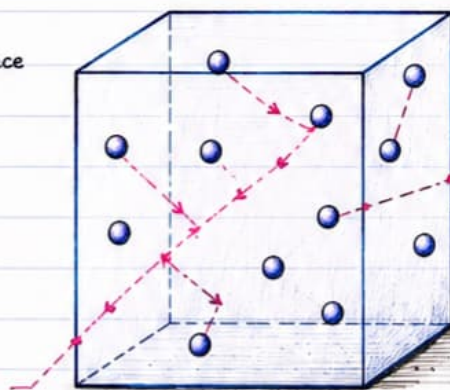


9. MEAN FREE PATH (λ)

- Mean free path (λ) is the average distance travelled by a molecule between two successive collisions.
- It depends on size of molecules, number density and temperature.
- For an ideal gas,

$$\lambda = \frac{1}{\sqrt{2} \pi d^2 n}$$

Where,

 d = diameter of molecule n = number density = $\frac{N}{V}$ 

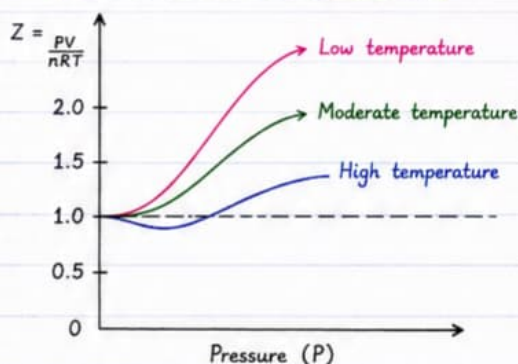
Path of a molecule

λ is large at low pressure and high temperature.

10. REAL GASES AND DEVIATION

- Real gases deviate from ideal behaviour at **high pressure** and **low temperature**.
- At high pressure $\rightarrow$ volume of molecules is not negligible.
- At low temperature $\rightarrow$ **intermolecular forces** become significant.
- At STP, most gases behave nearly as ideal gases.

DEVIATION FROM IDEAL BEHAVIOUR



- $Z = 1 \rightarrow$ Ideal gas behaviour
- $Z > 1 \rightarrow$ Positive deviation
- $Z < 1 \rightarrow$ Negative deviation

11. EXAMPLE

A gas at 300 K has a root mean square speed of 500 m s^{-1} . Find the molar mass of the gas.

Solution:

We know,

$$C_{rms} = \sqrt{\frac{3RT}{M}}$$

$$\Rightarrow M = \frac{3RT}{C_{rms}^2}$$

Substitute values:

$$\begin{aligned} M &= \frac{3 \times 8.314 \times 300}{(500)^2} \\ &= \frac{24.942 \times 10^3}{2.5 \times 10^5} \\ &= 0.0998 \text{ kg mol}^{-1} \\ &= 99.8 \text{ g mol}^{-1} \end{aligned}$$

Given:

$$T = 300 \text{ K}$$

$$C_{rms} = 500 \text{ m s}^{-1}$$

$$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$$

Find: $M = ?$

Higher temperature means higher molecular speed.

12. SUMMARY

- Gases consist of a very large number of tiny molecules in continuous **random motion**.
- Collisions between molecules and with container walls are **perfectly elastic**.
- Pressure is due to these collisions.
It is given by $P = \frac{1}{3} \rho \bar{c}^2$
- Average kinetic energy of a molecule = $\frac{1}{2} m \bar{c}^2 = \frac{3}{2} kT$
- Pressure of an ideal gas: $PV = nRT$
- Mean free path: $\lambda = \frac{1}{\sqrt{2} \pi d^2 n}$
- Root mean square speed: $C_{rms} = \sqrt{\frac{3RT}{M}}$
- Kinetic theory is valid for ideal gases (low pressure, high temperature).
- Pressure $\propto$ **Temperature** (at constant volume).

Key Formulas

$$P = \frac{1}{3} \rho \bar{c}^2$$

$$\frac{1}{2} m \bar{c}^2 = \frac{3}{2} kT$$

$$\hat{P}V = nRT$$

$$C_{rms} = \sqrt{\frac{3RT}{M}}$$

$$\lambda = \frac{1}{\sqrt{2} \pi d^2 n}$$

Remember: Kinetic theory connects the microscopic world of molecules with the macroscopic properties of gases.